



Presentation Topic: UNIX Operating System

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WHAT IS UNIX

UNIX is a powerful, multi-user, multitasking operating system.

Developed at Bell Labs by Ken Thompson & Dennis Ritchie in 1969.

Written in C language, making it portable across different hardware.

Foundation for modern OSes like Linux, macOS, Android, and iOS.

Widely used in servers, workstations, and supercomputers worldwide.



```
$ uname -a  
Linux unix-os 5.15.0  
GNU/Linux x86_64
```



Born: Bell Labs, 1969

Language: C

Family: POSIX / Unix-like

"The UNIX philosophy: Write programs that do one thing and do it well."

HISTORY OF UNIX

1969

Ken Thompson & Dennis Ritchie develop UNIX at Bell Labs (AT&T).

1983

Richard Stallman launches GNU Project (free UNIX-like software).

1973

UNIX rewritten in C language — first portable OS.

1991

Linus Torvalds creates Linux kernel, inspired by UNIX.

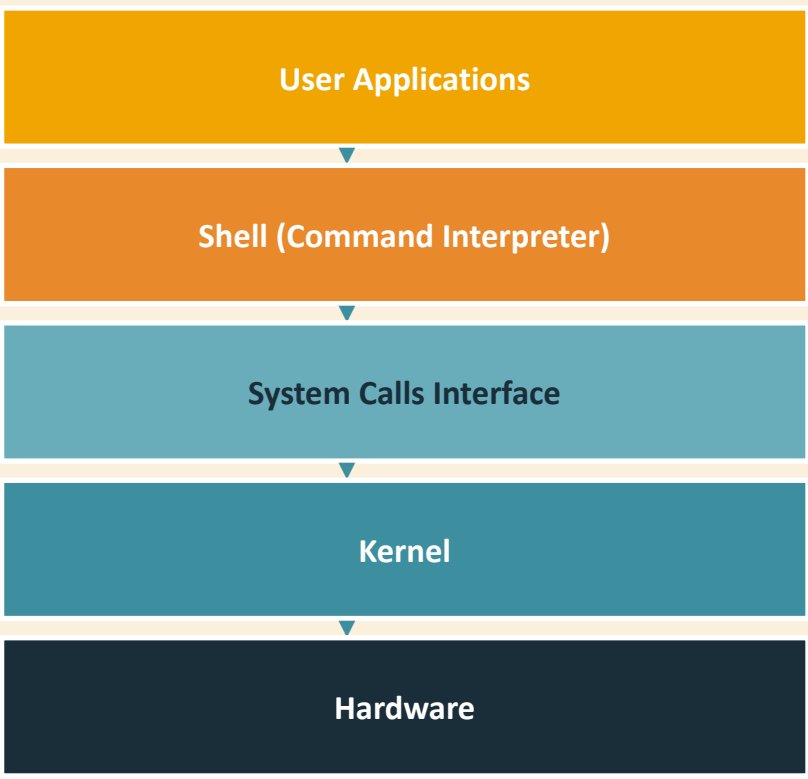
1977

BSD (Berkeley Software Distribution) released by UC Berkeley.

2000s

UNIX influences macOS, Android, iOS, and cloud infrastructure.

UNIX ARCHITECTURE



```
graph TD; A[User Applications] --> B[Shell (Command Interpreter)]; B --> C[System Calls Interface]; C --> D[Kernel]; D --> E[Hardware];
```

User Applications

Shell (Command Interpreter)

System Calls Interface

Kernel

Hardware

Key Components:

Kernel: Core of UNIX. Manages CPU, memory, devices, and processes.

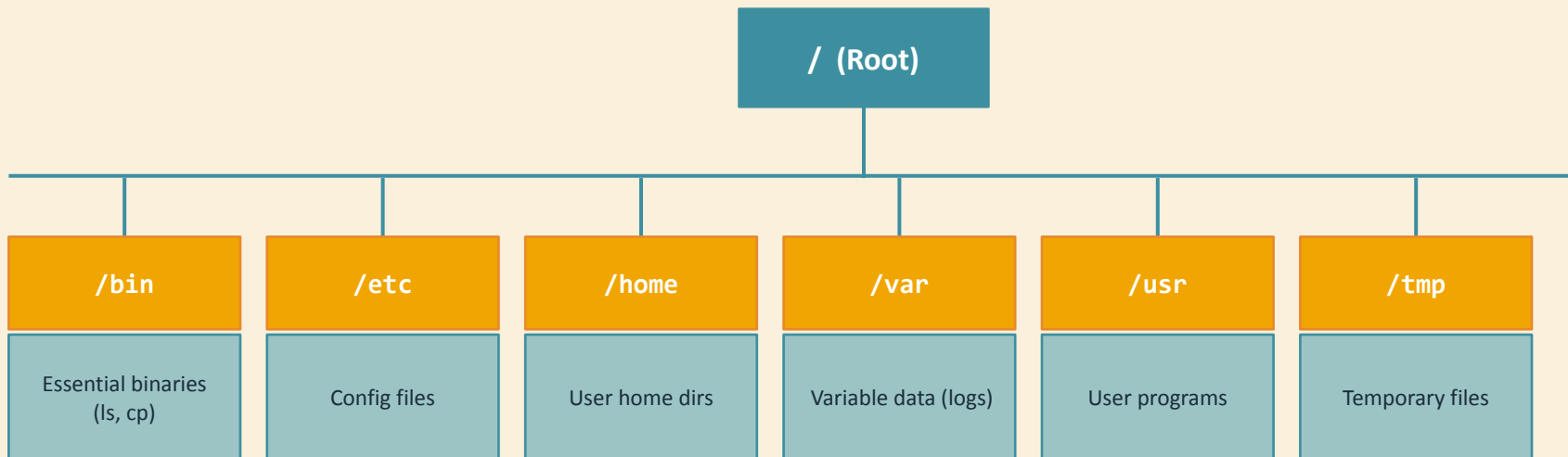
Shell: Interface between user and kernel. Interprets commands (bash, sh, csh).

System Calls: Programs communicate with the kernel via system calls.

File System: Hierarchical structure starting from root '/' directory.

Utilities: Built-in commands like ls, cp, grep, awk, sed.

UNIX FILE SYSTEM



Everything is a file — devices, directories, and processes are treated as files in UNIX.

Hierarchical structure — single-root tree starting from '/' (root directory).

Permissions system — each file has read (r), write (w), execute (x) for owner, group, others.

UNIX SHELL & COMMANDS

Common Shells:

- sh** — Bourne Shell (original)
- bash** — Bourne Again Shell (most popular)
- cs**h — C Shell
- ksh** — Korn Shell
- zsh** — Z Shell (modern, feature-rich)



Pro Tip — Pipes & Redirection:

```
ls -l | grep '.txt' | wc -l  
> output.txt (redirect to file)  
< input.txt (input from file)
```

Essential Commands:

ls

List directory

cd

Change directory

pwd

Print work dir

cp

Copy files

mv

Move/rename

rm

Remove files

grep

Search text

chmod

Change permissions

ps

List processes

kill

Terminate process

man

Manual pages

pipe |

Chain commands

UNIX PROCESSES & MULTITASKING

Process: A running instance of a program. Each has a unique PID (Process ID).

Multitasking: UNIX can run many processes simultaneously using time-sharing.

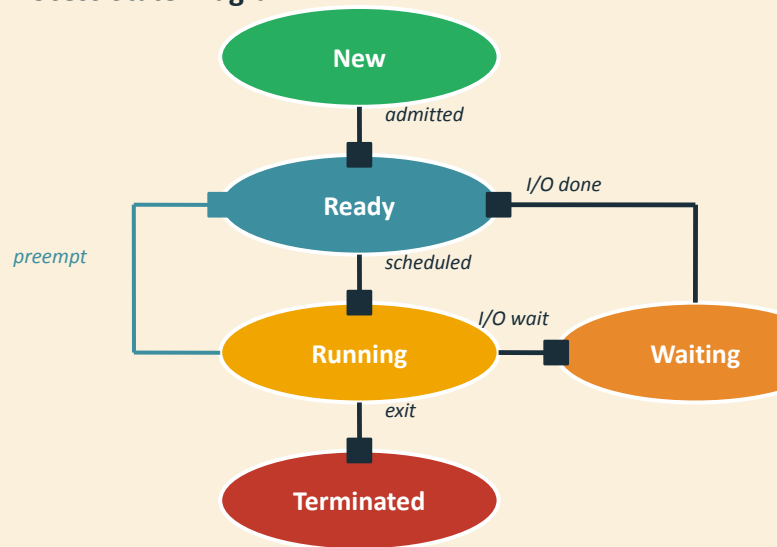
Foreground/Background: Processes can run in the foreground or background (&).

fork() & exec(): New processes created by forking a parent process.

Daemon processes: Background services (e.g., httpd, sshd) that run silently.

kill / signals: Processes communicate via signals (SIGKILL, SIGTERM, SIGHUP).

Process State Diagram:



UNIX SECURITY & FILE PERMISSIONS

Permission String Example:

-

Type
(file/dir)

rwX

Owner
(read/write/exec)

r-x

Group
(read/exec)

r--

Others
(read only)

chmod Commands:

```
chmod 755 file → rwxr-xr-x
chmod 644 file → rw-r--r--
chmod +x file → adds execute
chown user file → change owner
sudo → run as superuser
```

Security Features:

Multi-user isolation: Each user has private files.
root/superuser: Full system control.
SSH: Secure remote login.
PAM: Pluggable authentication.
Audit logs: Track user activity.

UNIX VS LINUX — KEY DIFFERENCES

Feature	UNIX	Linux
Origin	Bell Labs, 1969	Linus Torvalds, 1991
License	Proprietary (mostly)	Open source (GPL)
Cost	Expensive	Free to use
Source Code	Closed (mostly)	Fully open
Portability	Limited hardware	Wide hardware support
Examples	Solaris, HP-UX, AIX	Ubuntu, Fedora, Debian
Users	Enterprises, servers	Everyone — servers to desktops

ADVANTAGES OF UNIX



Highly
Secure



Multi-User
Support



Efficient
Memory Use



Built-in
Networking



Stable &
Reliable



Powerful
Tools

UNIX IN THE REAL WORLD

Web Servers: Apache and Nginx run on UNIX/Linux servers powering over 70% of the internet.

Supercomputers: 99%+ of the world's top 500 supercomputers run Linux (UNIX-based).

Cloud Computing: AWS, Google Cloud, and Azure rely on Linux/UNIX for infrastructure.

macOS & iOS: Apple's macOS and iOS are built on a UNIX foundation (Darwin/BSD).

Android: Android uses the Linux kernel — a direct UNIX descendant.

Banking & Finance: Core banking systems and ATMs often run on UNIX (AIX, Solaris).

Space & Military: NASA, SpaceX, and defense systems use UNIX-based systems for reliability.

SUMMARY



UNIX was born at Bell Labs in 1969 — the ancestor of modern operating systems.



Built on a kernel, shell, and hierarchical file system architecture.



Everything is a file — devices, processes, and data all follow a unified model.



Strong security via user permissions, root access, and SSH.



Powerful multitasking — fork, exec, daemons, and signals.



Foundation of Linux, macOS, Android, and the cloud infrastructure of today.

THANK YOU

Questions & Discussion

